

Supplementary Information IV

Supplementary Methods IV

[Supplementary methods describing architecture tuning and model selection to be completed from the final scripts and logs]

Supplementary Figures IV

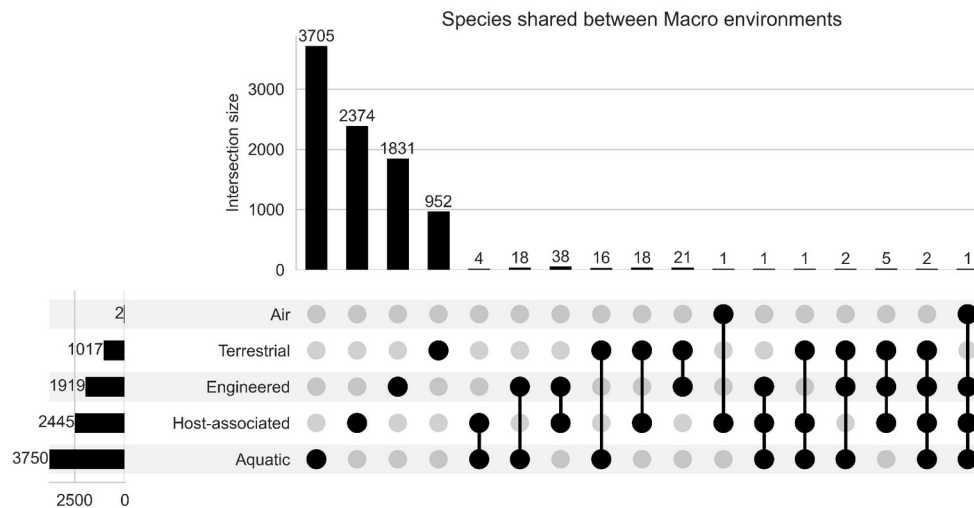


Figure IV-S1. Distribution of species-level representatives across combinations of macro-environmental labels. Bars show the number of representatives associated with each label combination, while the matrix indicates the macro-environments included in each intersection. Representatives associated with more than one macro-environment were excluded from model development and evaluated separately.

[Figure IV-S2 to be inserted]

Figure IV-S2. Moran's I calculated across neighbourhood sizes (k) for the weighted-mean and complete concatenated Evo 2 1B representations. Values are shown separately for the four macro-environments.

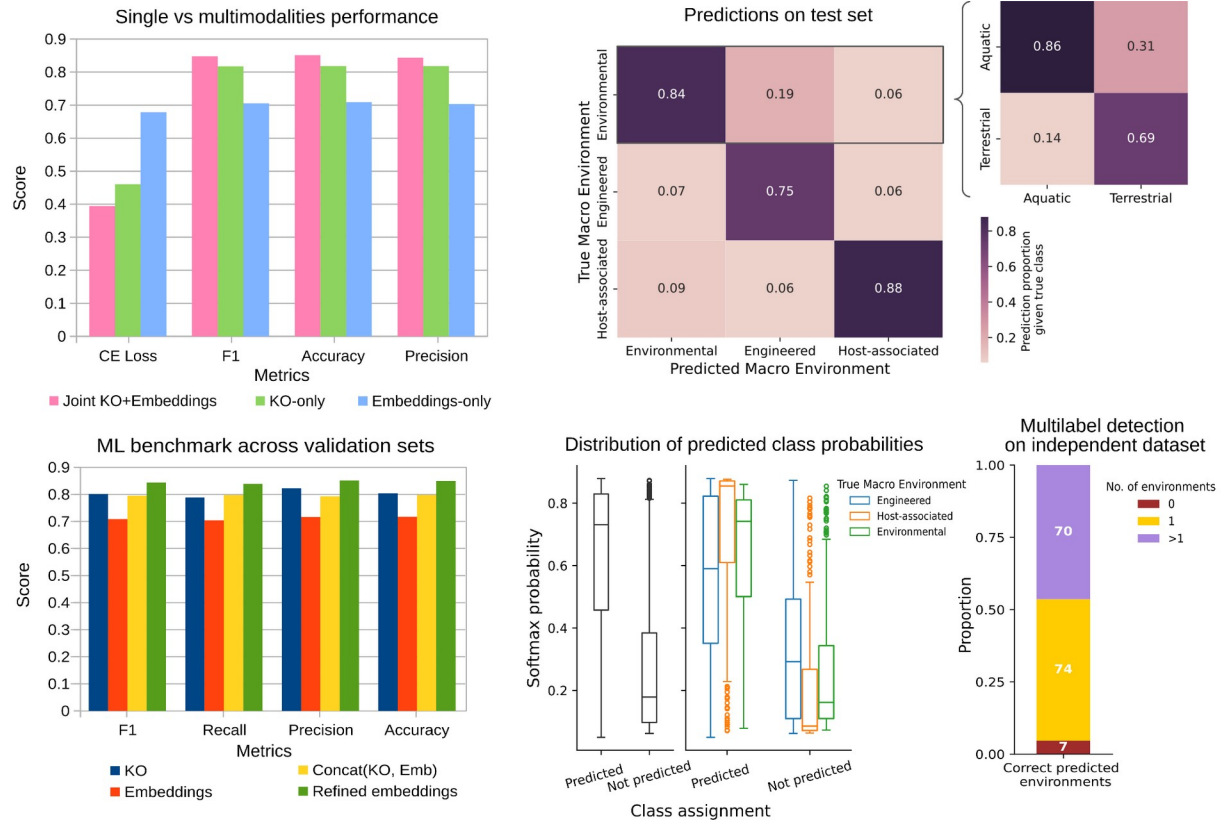
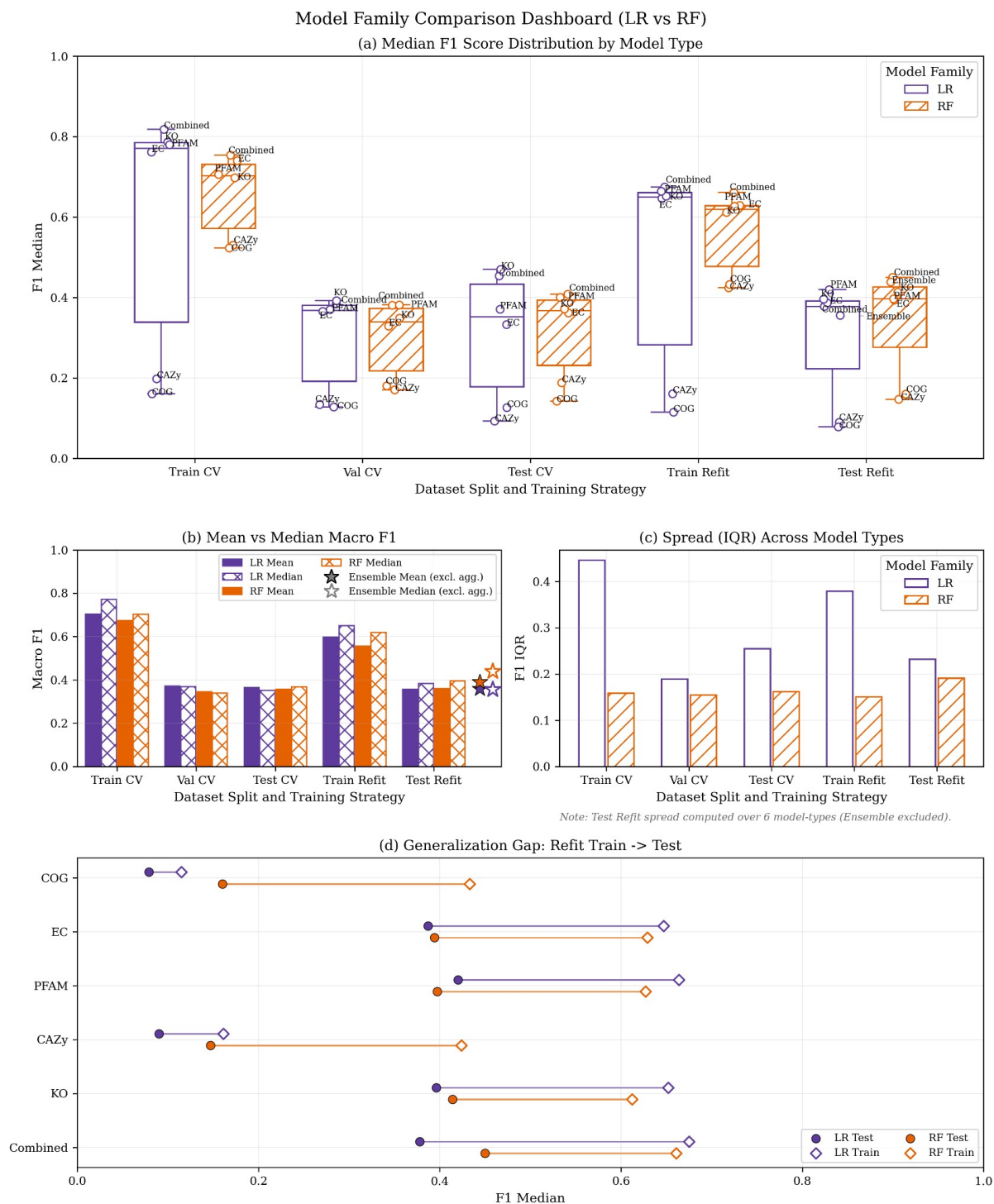


Figure IV-S3. Additional evaluation of macro-environment prediction. (A) Test-set cross-entropy loss, macro F1, accuracy and macro precision obtained using the integrated KO and Evo 2 representations or either modality alone. (B) Test-set performance of conventional ML benchmarks trained on KO profiles, Evo 2 embeddings, their direct concatenation or the refined multimodal representation. (C) Distribution of maximum softmax probabilities for correct and incorrect test-set predictions, shown separately for the true macro-environments. (D) Exploratory evaluation of the 151 multi-environment representatives according to the number of recorded macro-environments recovered among the highest-ranked predictions.



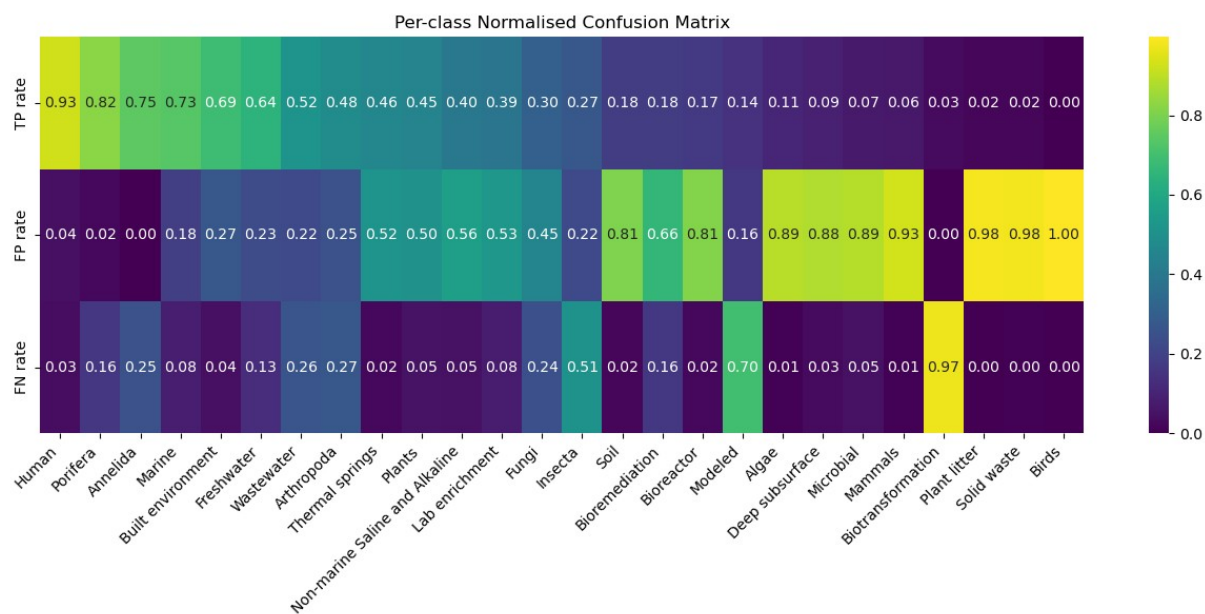


Figure IV-S5. Class-specific true-positive, false-positive and false-negative rates for the selected sub-environment model on the held-out test set. Environmental categories are ordered by decreasing true-positive rate. [Selected annotation system and classifier to be confirmed.]